

AI Dynamic Pricing Decision Engine

Overview

The EventRadius AI Dynamic Pricing Engine helps organizers make optimal pricing decisions to maximize revenue and fill perishable inventory (seats, slots, tables).

Core Principle: "AI recommends the best price at the best time to the best customer"

How AI Makes Pricing Decisions

Input Factors (Data Points)

Factor	Weight	Source
Current Occupancy	30%	Real-time booking data
Time to Event	25%	Event start time vs now
Historical Demand	20%	Past similar events performance
Day/Time Patterns	15%	Weekday vs weekend, time of day
External Factors	10%	Weather, holidays, local events

Decision Process

Step 1: Calculate Baseline Price

↓

Step 2: Apply Occupancy Adjustment

↓

Step 3: Apply Time Pressure Adjustment

↓

Step 4: Apply Demand Forecasting

↓

Step 5: Generate Final Recommendation

Pricing Decision Matrix

Decision 1: Should We Discount?

Current Occupancy	Time Remaining	AI Decision
>80%	>24 hours	❌ No discount needed
60-80%	>24 hours	❌ Monitor, no action
40-60%	12-24 hours	✅ Slight discount (10-15%)

Current Occupancy	Time Remaining	AI Decision
20-40%	6-12 hours	✔ Moderate discount (20-30%)
<20%	<6 hours	✔ Aggressive discount (35-50%)
<10%	<2 hours	✔ Maximum discount (50-60%)

Decision 2: How Much to Discount?

Base Price: \$20 (regular ticket price)

Scenario A: 65% sold, 18 hours remaining
→ AI recommends: \$18 (10% off)
→ Reason: Slight nudge to reach 85% capacity

Scenario B: 35% sold, 4 hours remaining
→ AI recommends: \$14 (30% off)
→ Reason: Fill seats that would be empty

Scenario C: 8% sold, 45 minutes remaining
→ AI recommends: \$10 (50% off)
→ Reason: Last-minute deal, any revenue better than \$0

Decision 3: Who to Target?

Discount Level	Target Audience	Notification Channel
10-15%	Past attendees, nearby users	Email, in-app
20-30%	Interest-matched, price-sensitive	Push notification
35-50%	Broad local audience	Push + SMS
50-60%	Everyone within radius	All channels + social

 AI Recommendation Examples

Example 1: Cinema Tuesday 7pm

Event: Movie showing, 100 seats **Current State:** 2 hours before showtime

Time	Occupancy	AI Recommendation	Action
4:00 PM (T-3h)	45%	"Monitor, no discount yet"	Wait
5:00 PM (T-2h)	48%	"Recommend 20% discount to nearby movie fans"	Send push
5:30 PM (T-1.5h)	52%	"Increase to 30% discount, expand radius"	Broaden targeting

Time	Occupancy	AI Recommendation	Action
6:00 PM (T-1h)	58%	"Final push: 40% off + free popcorn upsell"	Urgent alert
6:30 PM (T-30min)	70%	"Success: 18 additional seats filled"	Stop promoting

Result: 18 seats × \$12 avg = \$216 additional revenue (vs \$0 if empty)

Example 2: Yoga Studio Saturday Morning

Event: 9am Vinyasa class, 20 mats **Current State:** Friday evening

Metric	Value
Regular price	\$25
Bookings (Fri 6pm)	8/20 (40%)
Historical Sat 9am	Usually 90% full
Weather forecast	Sunny (good for attendance)

AI Analysis:

- Current occupancy (40%) is below historical average (90%)
- 15 hours remaining = time for recovery
- Sunny weather = positive factor
- Weekend morning = high demand time

AI Recommendation:

"Don't discount yet. Send reminder to:

- 12 past attendees who haven't booked
- 8 users with 'yoga' interest within 2 miles

Expected fill rate: 85% by 8am tomorrow"

Result: 17/20 filled at full price = \$425 revenue (no discount needed)

Example 3: Restaurant Tuesday Lunch

Event: 12pm-2pm lunch service, 30 tables **Current State:** Tuesday 10am

Metric	Value
Regular avg check	\$35
Reservations (10am)	8/30 tables (27%)
Historical Tue lunch	Usually 50% full

Metric	Value
Weather	Rainy (negative factor)
Local events	Convention downtown (positive)

AI Analysis:

- Current occupancy (27%) concerning
- Rainy weather = negative
- Convention nearby = positive offset
- Only 2 hours to service = urgency

AI Recommendation:

```
"Launch 'Rainy Day Lunch Special':
- 30% off all tables
- Target: Convention attendees (1 mile radius)
- Message: 'Escape the rain, warm lunch deal'
- Channel: Push notification to nearby users"
```

Pricing: \$35 → \$24.50 **Result:** 22 tables filled = \$539 revenue (vs \$280 without AI) **Gain:** +\$259 (+92% improvement)

AI Decision Framework

The 3 Questions AI Asks

1. WILL IT SELL?
 - Demand forecasting model
 - Predicts probability of sale at current price
 - If <30% probability: recommend discount
2. AT WHAT PRICE?
 - Price elasticity model
 - Finds optimal price for maximum revenue
 - Balances: discount depth vs fill rate
3. TO WHOM?
 - Audience targeting model
 - Matches event to most likely buyers
 - Prioritizes: past customers > interest match > location

AI Models Used

Model 1: Demand Forecasting

Purpose: Predict how many seats will sell at current price

```
# Simplified logic
def forecast_demand(occupancy, time_remaining, historical_avg):
    base_probability = occupancy / 100
    time_pressure = 1 / (time_remaining / 24) # Hours to days
    historical_trend = historical_avg / 100

    sale_probability = (base_probability * 0.4 +
                        time_pressure * 0.3 +
                        historical_trend * 0.3)

    return sale_probability
```

Output: "87% chance of selling at current price" or "23% chance - recommend discount"

Model 2: Price Optimization

Purpose: Find the price that maximizes expected revenue

```
# Expected Revenue = Price × Probability of Sale

def optimize_price(base_price, occupancy, time_remaining):
    test_discounts = [0, 0.10, 0.20, 0.30, 0.40, 0.50]
    best_revenue = 0
    best_discount = 0

    for discount in test_discounts:
        price = base_price * (1 - discount)
        probability = calculate_sale_probability(occupancy, time_remaining,
        discount)
        expected_revenue = price * probability

        if expected_revenue > best_revenue:
            best_revenue = expected_revenue
            best_discount = discount

    return best_discount, best_revenue
```

Example:

```
$20 ticket, 30% occupancy, 4 hours remaining:
- 0% discount: $20 × 15% chance = $3.00 expected
- 20% discount: $16 × 45% chance = $7.20 expected ← OPTIMAL
- 40% discount: $12 × 70% chance = $8.40 expected
```

AI recommends: 20% discount (best risk/reward)

Model 3: Audience Targeting

Purpose: Find the users most likely to book at the recommended price

Scoring Algorithm:

User Score =
(Interest Match × 0.3) +
(Distance Score × 0.25) +
(Past Behavior × 0.25) +
(Price Sensitivity × 0.2)

Where:
Interest Match: % overlap with event category
Distance Score: 1 – (miles / max_radius)
Past Behavior: Historical booking rate for similar events
Price Sensitivity: Response rate to past discounts

Result: Ranked list of users to notify

 Organizer Control Panel

Organizers can set AI parameters:

Setting 1: Discount Range

Option	Description	Use Case
Conservative	Max 20% discount	Premium brands, luxury venues
Balanced	Max 40% discount	Most venues (default)
Aggressive	Max 60% discount	High volume, budget venues
Custom	Set your own max	Specific brand requirements

Setting 2: Minimum Price Floor

"Never discount below \$X"

Example: \$20 ticket, \$12 floor

- AI can recommend: \$20, \$18, \$16, \$14, \$12
- AI cannot recommend: \$10, \$8, etc.

Setting 3: Auto-Pilot vs Manual

Mode	AI Action	Organizer Action
Auto-Pilot	Implements recommendations automatically	Monitors only
Assistant	Shows recommendations, suggests actions	Approves each action
Manual	Provides data and insights only	Makes all decisions

Setting 4: Notification Preferences

Setting	Options
Target Radius	1 mile, 3 miles, 5 miles, 10 miles
User Types	Past customers, Interest match, New users, Everyone
Channels	Push, Email, SMS, All
Frequency	Every discount, Daily summary, Only urgent

 AI Performance Metrics

Metrics for Organizers

Metric	Definition	Target
Fill Rate Improvement	% increase in occupancy vs without AI	+25%
Revenue Per Seat	Average revenue per available slot	Maximize
Discount Efficiency	% of discounts that resulted in sales	>70%
Customer Acquisition	New customers from AI promotions	Track growth
Return on AI	Additional revenue / Commission paid	>10x

Metrics for EventRadius

Metric	Purpose
Recommendation Accuracy	% of AI recommendations that filled seats
Price Elasticity Accuracy	How close predicted vs actual sales
User Engagement Rate	% of notified users who viewed event
Conversion Rate	% of views that resulted in booking
Organizer Satisfaction	Feedback on AI usefulness

 Continuous Learning Loop

1. AI Makes Recommendation
↓
2. Organizer Implements (or approves)
↓
3. Users Receive Notification
↓
4. Some Users Book (conversion)
↓
5. AI Measures Results
↓
6. AI Updates Models
↓
7. Future Recommendations Improve

Learning Sources:

- Which discounts worked? Which didn't?
- Which users responded? Which ignored?
- What times/days were most effective?
- What messaging drove conversions?

AI Decision Examples by Industry

Cinema Chain

Scenario: Tuesday 3pm showing (traditionally empty)

AI Decision Process:

Input:

- Time: 3 hours before showing
- Occupancy: 15% (15/100 seats)
- Historical: Tuesday 3pm usually 20% full
- Weather: Rainy (positive for movie attendance)
- Local: School holiday (positive)

Analysis:

- Occupancy is below historical
- Time pressure is high (3 hours)
- External factors positive (rain + holiday)
- Opportunity for significant discount

Recommendation:

- Discount: 50% off (\$12 → \$6)
- Target: Families with kids (school holiday)
- Message: "Rainy day movie special! \$6 tickets"
- Channel: Push to parents within 5 miles

Expected Outcome:

- Fill 40 additional seats
- Revenue: $40 \times \$6 = \240
- Without AI: $15 \text{ seats} \times \$12 = \$180$
- Net gain: $+\$60 + 25 \text{ more customers}$

Fitness Studio

Scenario: Wednesday 6pm spin class

AI Decision Process:

Input:

- Time: 18 hours before class
- Occupancy: 60% (12/20 bikes)
- Historical: Wed 6pm usually 95% full
- Weather: Sunny (negative – people exercise outside)
- Trend: Booking rate dropping

Analysis:

- Occupancy is concerning for this time slot
- Sunny weather is negative factor
- Still time to recover (18 hours)
- Don't panic discount yet

Recommendation:

- Discount: 15% off ($\$25 \rightarrow \21.25)
- Target: Past attendees who haven't booked
- Message: "Your usual Wednesday spot is waiting"
- Channel: Email + push to loyal customers

Expected Outcome:

- Fill 5 additional bikes
- Revenue: $5 \times \$21.25 = \106.25
- Without AI: likely 0–2 more sales
- Net gain: $+\$80\text{--}100$

Restaurant

Scenario: Thursday 2:30pm table (off-peak)

AI Decision Process:

Input:

- Time: 1 hour before service
- Reservations: 25% (10/40 tables)
- Historical: Thursday lunch usually 60% full
- Local: Business district, office workers nearby

- Weather: Nice (people might go out)

Analysis:

- Occupancy is low for time remaining
- Office workers are ideal target
- Short window (1 hour)
- Need aggressive discount

Recommendation:

- Discount: 35% off lunch menu
- Target: Office workers within 0.5 miles
- Message: "Quick lunch break deal – 35% off next hour"
- Channel: Push notification (immediate)

Expected Outcome:

- Fill 18 additional tables
- Revenue: $18 \times \$22 \text{ avg} = \396
- Without AI: $10 \text{ tables} \times \$35 = \$350$
- Net gain: $+\$46 + 8 \text{ more customers}$

AI Training Data

What Data Trains the Models?

Data Source	Weight	Privacy
Historical event performance	40%	Aggregated, anonymized
Booking patterns by time/day	25%	Anonymized
User preferences & interests	20%	Opt-in only, encrypted
Pricing response data	15%	Aggregated trends

How Organizers Benefit from Collective Intelligence

Yoga Studio A shares data:

- "25% discount fills 40% more seats on rainy days"

AI learns:

- Rainy weather + fitness classes = higher discount sensitivity

Yoga Studio B benefits:

- AI applies this learning to their rainy day pricing
- Better results without experimentation

AI Features Roadmap

Phase 1 (Current): Reactive Pricing

- Respond to low occupancy with discounts
- Basic demand forecasting
- Simple audience targeting

Phase 2: Predictive Pricing

- Predict low demand before it happens
- Preemptive pricing adjustments
- Advanced audience matching

Phase 3: Prescriptive Pricing

- AI suggests optimal event times
- Recommends best days for maximum revenue
- Predicts ideal capacity limits

Phase 4: Autonomous Pricing

- Fully automated pricing (organizer sets guardrails)
- Real-time price adjustments
- Cross-event optimization

 Organizer AI Dashboard

Real-Time View

 Cinema: Avengers – Tuesday 7pm

 Current Status:

- 45/100 seats filled (45%)
- 3 hours to showtime
- \$12 regular price

 AI Recommendation:

- Suggest 25% discount (\$9)
- Target: 20 nearby movie fans
- Expected: +15 seats, +\$135

 Approve & Send

 Adjust

 Skip

Historical Performance

 AI Performance This Month

- 23 AI recommendations made
- 18 resulted in sales (78% success)
- \$3,240 additional revenue generated
- \$97 commission paid to EventRadius
- ROI: 33.4x return on AI commission

Frequently Asked Questions

Q: Will AI force me to discount?

A: No. AI provides recommendations. You choose whether to implement.

Q: Can I override AI suggestions?

A: Yes. You can adjust discount amount, target audience, or skip entirely.

Q: What if AI is wrong?

A: You only pay commission on actual sales. Bad recommendations cost you nothing.

Q: Does AI learn my specific venue?

A: Yes. AI improves as it learns your customer patterns, preferences, and response rates.

Q: Can I set maximum discount limits?

A: Yes. Set price floors and discount caps in your AI preferences.

Q: How often does AI make recommendations?

A: Continuously monitored. Recommendations come when conditions trigger them (low occupancy + time pressure).

Summary

AI Value Proposition:

"Our AI analyzes thousands of data points to recommend the exact price, timing, and audience that will maximize your revenue. You stay in control - AI is your smart assistant, not your boss."

Key Benefits:

1. **No guesswork:** Data-driven pricing decisions
2. **Higher revenue:** Fill seats that would be empty
3. **Time savings:** AI monitors 24/7, alerts when action needed
4. **Better customers:** Targeted promotions to ideal audience
5. **Learning system:** Gets smarter with every event

Result: Organizers make more money with less effort.

Document Version: 1.0 **Last Updated:** April 2026 **Owner:** Product & AI Team